Linear Algebra 2026 Homework 1

Due: September 25, 2026

Problems should be solved in English and calculation steps are required.

Problem 1 [10 pts]. The augmented matrix of a linear system has been reduced by row operations to the form shown below. Continue the appropriate row operations (till its reduced echelon form) and describe the solution set of the system.

$$\begin{bmatrix} 1 & -5 & 4 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix}$$

Problem 2 [10 pts]. Solve the following system of linear equations:

$$\begin{aligned} x_1 \quad\quad\quad -3x_3 &= 8 \\ 2x_1 + 2x_2 + 9x_3 &= 7 \\ x_2 + 5x_3 &= -2 \end{aligned}$$

Problem 3 [8 pts]. True or False. You don't need to justify your answer.

1. Two matrices are row equivalent if they have the same number of rows.
2. Elementary row operations on an augmented matrix never change the solution set of the associated linear system.
3. Two equivalent linear systems can have different solution sets.
4. A consistent system of linear equations has one or more solutions.

Problem 4 [8 pts]. For the following matrices, determine which matrices are in reduced echelon form and which others are only in echelon form.

a. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$

Problem 5 [10 pts]. Row reduce the matrix to reduced echelon form. Circle the pivot positions in the final matrix and in the original matrix, and list the pivot columns.

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

Problem 6 [5 pts]. Find the general solutions of the system whose augmented matrix is given below.

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$$

Problem 7 [10 pts]. Find the general solutions of the system whose augmented matrix is given below.

$$\begin{bmatrix} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{bmatrix}$$

Problem 8 [8 pts]. The following matrices are in echelon form. The leading entries (■) may have any nonzero value; the starred entries (∗) may have any value (including zero). Suppose each matrix represents the augmented matrix for a system of linear equations. In each case, determine if the system is consistent. If the system is consistent, determine if the solution is unique.

a. $\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 0 & \blacksquare & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & 0 \end{bmatrix}$

Problem 9 [6 pts]. Mark each statement True or False. You do not need to justify each answer.

a. In some cases, a matrix may be row reduced to more than one matrix in reduced echelon form, using different sequences of row operations.

b. A basic variable in a linear system is a variable that corresponds to a pivot column in the coefficient matrix.

c. If one row in an echelon form of an augmented matrix is $[0 \quad 0 \quad 0 \quad 5 \quad 0]$, then the associated linear system is inconsistent.

Problem 10 [10 pts]. Mark each statement True or False. You do not need to justify each answer.

a. The reduced echelon form of a matrix is unique.

b. If every column of an augmented matrix contains a pivot, then the corresponding system is consistent.

c. The pivot positions in a matrix depend on whether row interchanges are used in the row reduction process.

d. A general solution of a system is an explicit description of all solutions of the system.

e. Whenever a system has free variables, the solution set contains many solutions.

Problem 11 [4 pts]. Compute $\mathbf{u}+\mathbf{v}$ and $\mathbf{u}-2\mathbf{v}$.

$$\mathbf{u}=\begin{bmatrix}-1\\2\end{bmatrix},\quad \mathbf{v}=\begin{bmatrix}-3\\-1\end{bmatrix}$$

Problem 12 [4 pts]. Write a system of equations that is equivalent to the given vector equation.

$$x_1\begin{bmatrix}3\\-2\end{bmatrix}+x_2\begin{bmatrix}7\\3\end{bmatrix}+x_3\begin{bmatrix}-2\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

Problem 13 [4 pts]. Write a vector equation that is equivalent to the given system of equations.

$$\begin{aligned}x_2+5x_3&=0\\4x_1+6x_2-x_3&=0\\-x_1+3x_2-8x_3&=0\end{aligned}$$

Problem 14 [8 pts]. Determine if $\mathbf{b}$ is a linear combination of the vectors formed from the columns of the matrix A.

$$A=\begin{bmatrix}1&-4&2\\0&3&5\\-2&8&-4\end{bmatrix},\quad \mathbf{b}=\begin{bmatrix}3\\-7\\-3\end{bmatrix}$$

Problem 15 [10 pts]. Let $\mathbf{a}_1=\begin{bmatrix}1\\3\\-1\end{bmatrix}, \mathbf{a}_2=\begin{bmatrix}-5\\-8\\2\end{bmatrix}$, and $\mathbf{b}=\begin{bmatrix}3\\-5\\h\end{bmatrix}$. For what value(s) of h is $\mathbf{b}$ in the plane spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$?

Problem 16 [8 pts]. Let $\mathbf{u}=\begin{bmatrix}2\\-1\end{bmatrix}$ and $\mathbf{v}=\begin{bmatrix}2\\1\end{bmatrix}$. Show that $\begin{bmatrix}h\\k\end{bmatrix}$ is in Span $\{\mathbf{u},\mathbf{v}\}$ for all h and k.

Problem 17 [4 pts]. Use the vector $\mathbf{u}=(u_1,\ldots,u_n)$ to verify the following algebraic properties of $\mathbb{R}^n$.

a. $\mathbf{u}+(-\mathbf{u})=(-\mathbf{u})+\mathbf{u}=\mathbf{0}$

b. $c(d\mathbf{u})=(cd)\mathbf{u}$ for all scalars c and d

Problem 18 [8 pts]. Compute the following products using (a) the definition, as in Example 1, on page 35 of the book *Linear Algebra and its applications, Forth edition by David C. Lay*, and (b) the row-vector rule for computing $A\mathbf{x}$, as in Example 5, on page 38 of the book *Linear Algebra and its applications, Forth edition by David C. Lay.* If a product is undefined, explain why.

1.
$$\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix}$$

2.
$$\begin{bmatrix} 1 & 3 & -4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Problem 19 [4 pts]. Use the definition of $A\mathbf{x}$ to write the following vector equation as a matrix equation.

$$x_1 \begin{bmatrix} 4 \\ -1 \\ 7 \\ -4 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 3 \\ -5 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ -8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}$$

Problem 20 [4 pts]. Write the following system first as a vector equation and then as a matrix equation.

$$\begin{aligned} 5x_1 + x_2 - 3x_3 &= 8 \\ 2x_2 + 4x_3 &= 0 \end{aligned}$$

Problem 21 [10 pts]. Let $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \\ -3 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ -3 \\ 9 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 4 \\ -2 \\ -6 \end{bmatrix}$. Does $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ span $\mathbb{R}^3$? Why or why not?

Problem 22 [10 pts]. Mark each statement True or False. **Justify each answer.**

a. The equation $A\mathbf{x} = \mathbf{b}$ is referred to as a *vector equation*.

b. A vector $\mathbf{b}$ is a linear combination of the columns of a matrix A if and only if the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution.

c. The equation $A\mathbf{x} = \mathbf{b}$ is consistent if the augmented matrix $[A \quad \mathbf{b}]$ has a pivot position in every row.

d. If the columns of an $m \times n$ matrix A span $\mathbb{R}^m$, then the equation $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b}$ in $\mathbb{R}^m$.

e. If A is an $m \times n$ matrix and if the equation $A\mathbf{x} = \mathbf{b}$ is inconsistent for some $\mathbf{b}$ in $\mathbb{R}^m$, then A cannot have a pivot position in every row.